

Speech Enhancement Using Non-negative Matrix Factorization Solved By Improved Alternating Direction Method Of Multipliers

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Abstract—For the task of monaural speech enhancement, A version of Sparse Nonnegative matrix factorization (Sparse NMF) using improved Alternating Direction Method of Multipliers (IADMM) with generalized Kullback-Leibler divergence is proposed. In this paper, an alternating direction method of multipliers (ADMM) for NMF is studied, which deals with the NMF problem using the cost function of beta divergence. Our study shows that this algorithm outperforms state-of-the-art algorithms on synthetic data sets, but the study shows that it presents unstable behavior and low accuracy on real data sets. Therefore, we propose an improved algorithm for sparse NMF to solve this problem. The algorithm minimizes the K-L divergence with a pivot element weighting iterative (PEWI) method. Experimental results demonstrated that the proposed algorithm is more stable and accurate and also obtained better performance than the state-of-the-art speech enhancement algorithms.

Keywords—Speech enhancement; NMF; ADMM; PEWI

I. INTRODUCTION

With the vastly usage of smart phones, digital cameras and other video-recording devices, user-generated video and audio keeps growing rapidly on internet, among which speech data takes a large proportion. Therefore, content based speech retrieval will perform an important role in information retrieval. Currently, speech retrieval largely relies on an automatic speech recognizer (ASR) or operated by spectrum comparison. Both methods will deteriorate severely with the appearance of noise. Hence, speech enhancement is usually a necessary pre-processing step to purify the recorded noisy speech. This paper investigates the monaural speech enhancement problem, where speech was recorded by a single microphone.

Non-negative matrix factorization (NMF) has been an active research topic of great importance in signal processing over the decades. NMF has been a useful dimensionality reduction approach for extracting interpretable features from non-negative observations [1]. The model used in the NMF algorithm proposed by Lee and Seung is a bilinear factorization of a non-negative matrix input $\mathbf{V}$ into two non-

negative matrices $\mathbf{W}$ and $\mathbf{H}$ [2], i.e. matrices satisfying the following equation:

$$\mathbf{V} = \hat{\mathbf{V}} \approx \mathbf{WH} \quad (1)$$

The factorization model is conceptually similar to other matrix factorizations such as principal component analysis (PCA) which is commonly used in matrix factorizations. What makes the model particularly interesting is that the NMF algorithms constrain the matrices to be entry-wise nonnegative which makes it particular suitable for reflecting the additive nature of sound components from different mixtures.

The NMF do the decomposition by minimizing the reconstruction error between the observation matrix $\mathbf{V}$ and the model $\mathbf{WH}$ while respecting the non-negative nature of the observe data, i.e. to find a solution of the following optimization problem:

$$\begin{aligned} \min D(\mathbf{V} | \mathbf{WH}) \\ \text{s.t. } \mathbf{V} \in \mathbb{R}_{\geq 0}^{m \times n}, \mathbf{W} \in \mathbb{R}_{\geq 0}^{m \times r}, \mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times n} \end{aligned} \quad (2)$$

The beta divergence $D(\mathbf{V} | \hat{\mathbf{V}})$ refers to the particular measurement of divergence between $\mathbf{V}$ and its corresponding reconstruction $\hat{\mathbf{V}}$. The Euclidean distance, the generalized Kullback-Leibler (K-L) divergence and the Itakura-Saito (IS) divergence are three most commonly used measurements with NMF [3]. The above measures implicitly assume a certain noise distribution.

In practice, the optimization problem is convex in $\mathbf{W}$ and $\mathbf{H}$ separately. The multiplicative update method proposed by Lee and Seung is a good compromise between speed and ease of implementation for solving the optimization problem [4]. But the convergence of the algorithm to a stationary point has not yet been proven [5]. The algorithms rely on special properties of Euclidean distance and they do not generalize readily to other situation of the beta-divergence. The multiplicative update method is easy to implement. Since sparsity induced by the non-negativity constraint is often an

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important property in many applications [6]. Sun et al. introduced ADMM framework to the NMF with beta divergence, proposing an ADMM based update framework (Universal ADMM) which addresses the above-mentioned shortcomings and yields superior results to the multiplicative updates algorithm [7].

However, it's discovered that for real nonnegative data, e.g., speech spectrum, Universal ADMM results in lower performance than our expectation in terms of stability and accuracy. Through the analysis of experimental results, it can be found that this phenomenon happens due to the ill-condition of matrices while updating dictionary matrix $\mathbf{W}$ and activation matrix $\mathbf{H}$. Zhang et al. proposed a stable ADMM algorithm for NMF coupled with an efficient method PEWI to solve this problem [8]. But the accuracy performance with this version of the stable ADMM algorithm on real non-negative data sets is still far from that on synthetic data sets due to the non-stationary property of the real spectrum data.

The motivation of this work is thus to devise an algorithm that addresses the issues mentioned above. We design an algorithm solving the problem of the sparse NMF with the cost function of KL divergence[9]. KL divergence is chosen as the cost function since it is more sensitive to low-energy observations than the Euclidean distance which makes it a better approximation of human auditory perception[10]. Moreover, we introduce the PEWI method into the universal ADMM to solve the ill-condition problem and improve the performance of the speech enhancement algorithm [11].

II. RELATED WORKS

Sparsity is a useful property of NMF as it usually produces a sparse representation of the data. The non-negativity constraints make the representation purely additive (allowing no subtractions), in contrast to many other linear representations such as PCA [12] and independent component analysis(ICA) [13]. The sparsity pattern induced by the non-negativity constraint is often of interest in many applications [14]. Moreover, ADMM can yield sparse matrices (~80% of the estimated activations for the mixture signal are zeros). In this paper, we introduce sparse NMF with the cost function of KL divergence to utilize sparse property which is yielded by ADMM.

A. Sparse NMF with K-L Divergence

To address the aforementioned issues, we propose to optimize the following problem:

$$\begin{aligned} \min D_{KL}(\mathbf{V} | \mathbf{WH}) + \lambda \|\mathbf{H}\|_1 \\ \text{s.t. } \mathbf{V} \in \mathbb{R}_{\geq 0}^{m \times n}, \mathbf{W} \in \mathbb{R}_{\geq 0}^{m \times r}, \mathbf{H} \in \mathbb{R}_{\geq 0}^{r \times n} \end{aligned} \quad (3)$$

where λ is the parameter to control the sparsity of the matrix $\mathbf{H}$ by minimising the L1-norm of its elements which controls the trade-off between sparsity and accuracy of the reconstruction. The notation $D_{KL}(\cdot)$ is the generalized K-L divergence defined as:

$$D_{KL}(x | \hat{x}) = x \log \frac{x}{\hat{x}} - x + \hat{x} \quad (4)$$

B. ADMM for Sparse NMF with K-L Divergence

In the approach proposed by Sun and F'evotte [7], the sparse NMF problem can be rewritten as

$$\begin{aligned} \min D_{KL}(\mathbf{V} | \mathbf{X}) + \lambda \|\mathbf{H}\|_1 \\ \text{s.t. } \mathbf{X} = \mathbf{WH} \\ \mathbf{W} = \mathbf{W}_+, \mathbf{H} = \mathbf{H}_+ \\ \mathbf{W}_+ \in \mathbb{R}_{\geq 0}^{m \times r}, \mathbf{H}_+ \in \mathbb{R}_{\geq 0}^{r \times n} \end{aligned} \quad (5)$$

The augmented Lagrangian including 5 primal and 4 dual variables according to the corresponding is as follows:

$$\begin{aligned} L_\rho(\mathbf{X}, \mathbf{W}, \mathbf{H}, \mathbf{W}_+, \mathbf{H}_+, \boldsymbol{\alpha}_x, \boldsymbol{\alpha}_w, \boldsymbol{\alpha}_h) = \\ D_{KL}(\mathbf{V} | \mathbf{X}) + \langle \boldsymbol{\alpha}_x, \mathbf{X} - \mathbf{WH} \rangle + \frac{\rho}{2} \|\mathbf{X} - \mathbf{WH}\|_F^2 \\ + \langle \boldsymbol{\alpha}_w, \mathbf{W} - \mathbf{W}_+ \rangle + \frac{\rho}{2} \|\mathbf{W} - \mathbf{W}_+\|_F^2 \\ + \langle \boldsymbol{\alpha}_h, \mathbf{H} - \mathbf{H}_+ \rangle + \frac{\rho}{2} \|\mathbf{H} - \mathbf{H}_+\|_F^2 \\ + \lambda \|\mathbf{H}\|_1 \end{aligned} \quad (6)$$

The variable split strategy of ADMM ensures that the separate optimization of variables is equivalent to their joint optimization. The algorithm can be summarized as below:

Algorithm1 ADMM for Sparse NMF with K-L divergence

- 1: **Input:** $\mathbf{v}$
 - 2: **Initialization:** $\mathbf{x}, \mathbf{w}, \mathbf{h}, \mathbf{w}_+, \mathbf{h}_+, \boldsymbol{\alpha}_x, \boldsymbol{\alpha}_w, \boldsymbol{\alpha}_h, \rho, \text{iter}$
 - 3: **for** $i = 1 : \text{iter}$
 - 4: $\mathbf{W}^T \leftarrow (\mathbf{H}\mathbf{H}^T + \mathbf{I})^{-1} \left((\mathbf{H}\mathbf{x}^T + \mathbf{W}_+^T) + \frac{1}{\rho} (\mathbf{H}\boldsymbol{\alpha}_x^T - \boldsymbol{\alpha}_w^T) \right)$
 - 5: $\mathbf{H} \leftarrow (\mathbf{W}^T\mathbf{W} + \mathbf{I})^{-1} \left(\mathbf{W}^T\mathbf{x} + \mathbf{H}_+ + \frac{1}{\rho} (\mathbf{W}^T\boldsymbol{\alpha}_x - \boldsymbol{\alpha}_h + \lambda) \right)$
 - 6: $\mathbf{X} \leftarrow \arg \min_{\mathbf{X} \geq 0} D_{KL}(\mathbf{V} | \mathbf{X}) + \langle \boldsymbol{\alpha}_x, \mathbf{X} \rangle + \frac{\rho}{2} \|\mathbf{X} - \mathbf{WH}\|_F^2$
 - 7: $\mathbf{W}_+ \leftarrow \max \left(\mathbf{W} + \frac{1}{\rho} \boldsymbol{\alpha}_w, 0 \right)$
 - 8: $\mathbf{H}_+ \leftarrow \max \left(\mathbf{H} + \frac{1}{\rho} \boldsymbol{\alpha}_h, 0 \right)$
 - 9: $\boldsymbol{\alpha}_x \leftarrow \boldsymbol{\alpha}_x + \rho(\mathbf{X} - \mathbf{WH})$
 - 10: $\boldsymbol{\alpha}_w \leftarrow \boldsymbol{\alpha}_w + \rho(\mathbf{W} - \mathbf{W}_+)$
 - 11: $\boldsymbol{\alpha}_h \leftarrow \boldsymbol{\alpha}_h + \rho(\mathbf{H} - \mathbf{H}_+)$
 - 12: **end**
 - 13: **Output:** $\mathbf{W}_+, \mathbf{H}_+$
-

In algorithm 1, the majorization of $\mathbf{X}$ can be solved through taking the derivative of L_ρ with respect to $\mathbf{X}$ and set it to zero, i.e.:

$$\frac{\partial L_\rho}{\partial \mathbf{X}} = -\frac{\mathbf{V}}{\mathbf{X}} + 1 + \alpha_x + \rho(\mathbf{X} - \mathbf{WH}) = 0 \quad (7)$$

In the process of equation (7), a quadratic equation should be solved. From the equation, we can obtain one positive and one negative root. The positive one is what we need and is given as follows:

$$\mathbf{X} \leftarrow \frac{(\rho \mathbf{WH} - \alpha_x - 1) + \sqrt{(\rho \mathbf{WH} - \alpha_x - 1)^2 + 4\rho \mathbf{V}}}{2\rho} \quad (8)$$

III. IMPROVED ADMM FOR SPARSE NMF

A. Existing Problem

In Universal ADMM algorithm, updating dictionary matrix $\mathbf{W}$ and activation matrix $\mathbf{H}$ requires solving system $\mathbf{Ax} = \mathbf{b}$. In many cases, the system established by the real data is ill-conditioned, so this common method may yield unstable and imprecise results. During experiments on speech spectrum, it's observed that Universal ADMM results in unstable and low-accuracy solution shown in Fig. 1.

If the system is ill-conditioned, small changes of right-hand vector $\mathbf{b}$ or coefficient matrix $\mathbf{A}$ may result in a large change in the solution. The condition number $\text{cond}(\cdot)$ is used as a measure of the relative sensitivity of solution $\mathbf{x}$ to changes in right-hand vector $\mathbf{b}$ and coefficient matrix $\mathbf{A}$. The condition number of matrix $\mathbf{A}$ is defined as:

$$\text{cond}(\mathbf{A}) = \|\mathbf{A}^{-1}\| \cdot \|\mathbf{A}\| \quad (9)$$

If $\text{cond}(\mathbf{A})$ becomes large, the system is regarded as being ill-conditioned [14]. In practice, the result shows that the point of solution mutation coincides with large value of the condition number from the test about the condition number of the system.

For speech data, since the speech energy usually focuses on certain frequency bands, the column features of dictionary $\mathbf{W}$ which captures prototypical spectra will become similar, this will make the condition number of coefficient matrix very large and change significantly which have a direct relationship with ill-condition of the matrix [15].

B. Pivot Element Weighting Iterative Method

As discussed in the previous section, the instability of ADMM essentially owes to large value and mutation of the condition number of coefficient matrix. Thus we propose a method to solve the ill-condition of coefficient matrix using Pivot Element Weighting Iterating (PEWI) algorithm.

The form of pivot element weighting is as follows:

$$\mathbf{A} + \alpha \mathbf{E} \quad (10)$$

Where $\mathbf{A}$ is a positive definite square matrix, $\mathbf{E}$ is the

identity matrix, and α is a weighting value. It has been proven that if $\mathbf{A}$ is a symmetric positive definite matrix and $\alpha > 0$, then we have $\text{cond}(\mathbf{A} + \alpha \mathbf{E}) < \text{cond}(\mathbf{A})$ [15]. Therefore, we can rewrite the ill-conditioned system $\mathbf{Ax} = \mathbf{b}$ as $(\mathbf{A} + \alpha \mathbf{E})\mathbf{x} = \mathbf{b} + \alpha \mathbf{x}$ which guarantees the small condition number. Then we can construct the iterative formula

$$(\mathbf{A} + \alpha \mathbf{E})\mathbf{x}_{k+1} = \mathbf{b} + \alpha \mathbf{x}_k \quad (11)$$

Let $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{e}_k$, then

$$(\mathbf{A} + \alpha \mathbf{E})\mathbf{e}_k = \mathbf{b} + \alpha \mathbf{x}_k \quad (12)$$

If $\mathbf{e}_k \rightarrow 0$, and $\mathbf{x}_{k+1} \rightarrow$ optimal solution, the proof of iterative convergence can be found in [15].

IV. EXPERIMENT RESULTS

A. Convergence Performance of Universal ADMM

In this section, we compare the performance of the multiplicative updates and the Universal ADMM algorithm with different parameter selections. The value of cost function is considered to examine the performance of the Universal ADMM algorithm and the improved ADMM algorithm based Sparse NMF. The matrix to be factorized was the magnitude spectrogram of 5 dB noise mixture, which is synthesized using clean speech from NOIZEUS (including 5 male and 5 female) and noise from NOISEX-92 database. All signals are resampled to 8 kHz and we set the iteration number to $1e5$. We obtain the spectrogram of noisy speech by using a Hamming window of 512 points with a hopping size of 256 points.

We consider different values of using the cost function of KL divergence to test algorithm on speech data. Multiplicative updates for NMF with generalized K-L divergence is also considered for comparison as was shown in Fig 1.

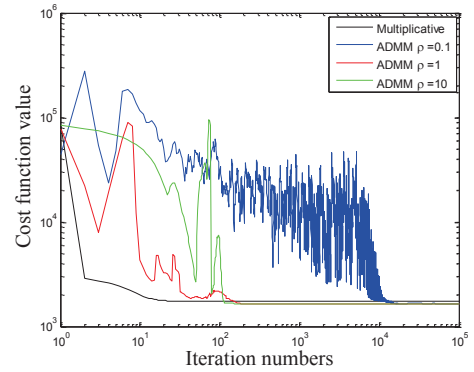


Fig. 1. Universal ADMM and Multiplicative updates performance on real spectrum data

The experiments show that for real speech spectrum, the Universal ADMM results in lower performance in terms of stability and accuracy. Especially when ρ is small, the convergence can be very unstable and have low accuracy. The parameter ρ seemed important to control the convergence rate of the proposed algorithm. A smaller value of ρ means a

larger step size which will finally lead to a faster convergence rate. However, since Universal ADMM in NMF does not guarantee monotonous decreasing as that in multiplicative updates [8], a smaller value of ρ may cause the fluctuation of the cost function values as was shown in the top panel in Fig.1. We can also observe from the bottom panel in Fig.1 that a larger value of ρ makes the convergence process more stable but also slowed down the convergence rate. A trade-off is required to balance the convergence speed and the stability of the optimization process.

B. Improved ADMM against Universal ADMM

The simulation results show the instability and low accuracy of Universal ADMM on real data sets. Therefore, we compare the performance of the improved ADMM with Universal ADMM for the task of speech enhancement. For the improved Algorithm, $\mathbf{X}_0$ is unit vector and the step size was chosen as $\alpha = 2$, which seemed to be able to make a good trade-off between stability of convergence and accuracy of solution.

The instability and low accuracy of have been shown through the experiments. The analysis found that they are caused by ill-conditional of linear of system on real data sets. To proposed one stable method: IADMM. The proposed algorithm attempts to solve the ill-condition problem using a pivot element weighting iterative method. As shown in Figure 2, the proposed algorithm achieves superior performance in both stability and accuracy compared to the Universal ADMM.

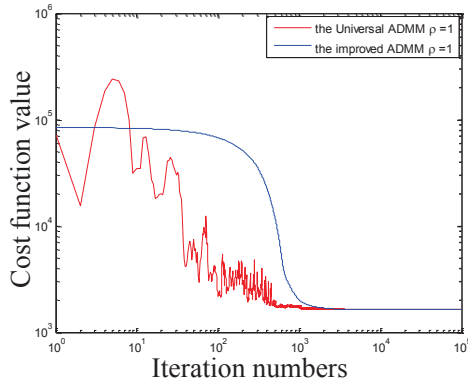


Fig. 2. Comparison of improved ADMM performance with Universal ADMM : K-L divergence as cost function and $\rho = 1$

The computational complexity is an important factor in concern. As was reported in Fig.2, the improved ADMM needs more iteration to reach a similar cost value as Universal ADMM does. Furthermore, the complexity per iteration is also higher than the Universal ADMM. However, IADMM can acquire a more stable and accurate approximation than Universal ADMM as was presented in Fig.2. Moreover, the yielded sparse matrix is really sparse with most of the elements are exactly zero other than a dense matrix full of tiny non-negative values as was obtained in multiplicative updates. So, the price of computational complexity is worth to pay once one has sufficient computing power.

C. Comparison with Universal ADMM and MSS

We compare the performance of IADMM based Sparse NMF with multiplicative updates for the task of speech separation. We compute the SDR of each algorithm after 2000 iterations. The results were reported in TABLE I:

TABLE I
SDR OF THE ESTIMATED SPEECH UNDER DIFFERENT INPUT SNR AND NOISE TYPES USING IADMM / MULTIPLICATIVE UPDATES

Noise Type	SNR			
	-5 dB	0 dB	5 dB	10 dB
M109	5.03/4.62	6.39/5.83	7.33/7.25	8.13/7.43
Volvo	2.83/2.28	7.94/8.33	10.51/10.79	10.75/10.81
Machinegun	9.05/8.51	9.36/8.93	9.89/9.70	13.72/12.29
Hfchannel	2.27/2.15	5.71/4.98	7.84/7.11	9.39/8.52
White	3.56/3.37	6.71/6.12	8.92/8.32	10.24/9.68

These results in Table I demonstrated that for most experiment settings the proposed algorithm archived superior performance than the multiplicative updates.

We also compare the proposed algorithm (IADMM) with two previous established approaches, the Universal ADMM (UADMM) and multi-band spectral subtraction (MSS) [16] on speech enhancement task using KL-divergence and ρ set to 1. All the methods were evaluated by using the BSS-EVAL metrics [17] and PESQ score [18]. BSS-EVAL is composed of three matrices, Source-to-Interference Ratio (SIR), Source-to-Artificial Ratio (SAR) and Source-to-Distortion Ratio (SDR). It is broadly used to evaluate the performance of signal separation. Larger value means better separation quality for all the three metrics. PESQ is another vastly used objective quality metric in speech enhancement. It is worth to note that, to make the comparisons fair, the proposed method and UADMM share the same parameters that control the sparsity of the spectrogram decomposition.

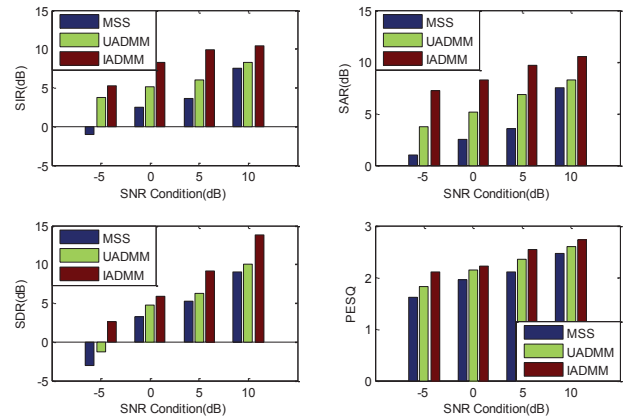


Fig. 3. Performance on SIR, SAR, SDR and PESQ

Figure 3 shows that the performance of the algorithms at different input SNR conditions. We can see that both UADMM and the proposed method obtained better results than MSS. When the PEWI algorithm was not applied, the UADMM algorithm still achieved comparable enhancement results with respect to MSS in terms of SAR (the metric that reflects the artifacts introduced by the separation process),

SDR, and PESQ. And the performance of the proposed algorithm performs better than the UADMM. However, the performance of the proposed algorithm on real non-negative data sets is still far from that on synthetic data sets.

V. CONCLUSION AND FUTURE WORK

We conducted a monaural speech enhancement by using an improved ADMM based NMF with generalized K-L divergence. In this paper, the instability and low accuracy of Universal ADMM have been shown through the experiments. The analysis found that they are caused by the ill-condition of linear system on real data sets. To solve the ill-condition problem, we proposed an improved ADMM using a pivot element weighting iterative method. Experimental results showed that the proposed algorithm yielded more stable, accurate solutions and performed better than MSS and UADMM and the multiplicative updates.

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REFERENCES

- [1] P. Smaragdis, C. Févotte, G. J. Mysore and N. Mohammadiha, "Static and dynamic source separation using nonnegative matrix factorizations: a unified view," *IEEE Signal Processing Magazine*, vol. 31, no. 3, pp. 66-75, 2014.
- [2] D. D. Lee and H. S. Seung, "Learning the parts of objects with nonnegative matrix factorization," *Nature*, vol. 401, pp. 788-791, 1999.
- [3] C. Févotte and J. Idier, "Algorithms for nonnegative matrix factorization with the β -divergence," *Neural Computation*, vol. 23, no. 9, pp. 2421-2456, 2011.
- [4] Lee D D. Algorithms for Non-negative Matrix Factorization[J]. *Advances in Neural Information Processing Systems*, 2001, 13(6):556--562.
- [5] Fortin M, Glowinski R. Augmented Lagrangian Methods: Application to the Numerical Solution of Boundary-Value Problems[J]. *Studies in Mathematics & Its Applications*, 1983, 15.
- [6] Hoyer P O. Non-negative Matrix Factorization with Sparseness Constraints[J]. *Neurocomputing*, 2012, 80(1):38-46.
- [7] D. L. Sun, C. Févotte, "Alternating direction method of multiplicative for non-negative matrix factorization with beta-divergence," in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2014, pp. 6201-6205.
- [8] S. F. Zhang, D. Huang, L. Xie, et al. Non-negative Matrix Factorization using Stable Alternating Direction Method of Multipliers for Source Separation[C]// *Apsipa Asc*. 2015.
- [9] S. Wang, N. Wang, D. Tao, et al. A K-L divergence constrained sparse NMF for hyperspectral signal unmixing[C]// *International Conference on Security, Pattern Analysis, and Cybernetics. IEEE*, 2014: págs. 6-18.
- [10] M. Sun, Y. Li, J. F. Gemmeke, et al. Speech Enhancement Under Low SNR Conditions Via Noise Estimation Using Sparse and Low-Rank NMF with Kullback-Leibler Divergence[J]. *Audio Speech & Language Processing IEEE/ACM Transactions on*, 2015, 23(7):1233-1242.
- [11] B. Stephen, P. Neal, C. Eric, P. Borja, & E. Joanthan. (2011). Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations & Trends® in Machine Learning*, 3(1), 1-122.
- [12] M. E. Davies and C. J. James, "Source separation using single channel ica," *Signal Processing*, vol. 87, no. 8, pp. 1819-1832, 2007.
- [13] F. Asano, Y. Motomura, H. Asoh, and T. Matsui, "Effect of pca filter in blind source separation," in *Proc. ICA*, 2000, pp. 57-62.
- [14] M. Shi and L. Gu, *Scientific and Engineering Calculation Bases*. Tsinghua University Press Ltd, 1999.
- [15] Tang L, Peng-Fei L I. A Pivot Element Weighting Iterative Method for Solving Ill-conditioned Linear Equations[J]. *Science Technology & Engineering*, 2012.
- [16] E. Vincent, R. Gribonval, and C. Févotte, "Performance measurement in blind audio source separation," *IEEE Trans. Audio, Speech, and Language Processing*, vol. 14, no. 4, pp. 1462-1469, 2006.
- [17] E. Vincent, R. Gribonval, and C. Févotte, "Performance measurement in blind audio source separation," *IEEE Trans. Audio, Speech, and Language Processing*, vol. 14, no. 4, pp. 1462-1469, 2006.
- [18] Hu Y, Loizou P C. Evaluation of Objective Quality Measures for Speech Enhancement[J]. *IEEE Transactions on Audio Speech & Language Processing*, 2008, 16(1):229-238.